

Dynamic Guideline: WHO-Based HIV - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This summary synthesizes the 5 most recent WHO HIV-related guidelines, with the latest update published 29 April 2026:

1. **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline
 - Publication Date: 29 April 2026
 - Target Population: Adolescents and youth (10-24 years) living with or at risk of HIV, HIV guideline development panels
 - Scope and Objectives: Establish standard requirements for youth representation in HIV guideline development, ensure recommendations align with the unique needs of youth populations
2. **Guideline Title:** WHO Consolidated Guidelines on HIV Prevention, Testing, Treatment, Service Delivery and Monitoring (2025 Update)
 - Publication Date: 15 January 2025
 - Target Population: All people living with HIV (PLHIV), populations at elevated HIV risk
 - Scope and Objectives: Update evidence-based recommendations across the full HIV care continuum to reduce global HIV incidence and mortality
3. **Guideline Title:** WHO Guidelines on the Management of Advanced HIV Disease (2024 Revision)
 - Publication Date: 22 August 2024
 - Target Population: PLHIV with CD4 count < 200 cells/mm³ or AIDS-defining illness
 - Scope and Objectives: Optimize care pathways for advanced HIV to reduce preventable mortality in low- and middle-income countries (LMICs)
4. **Guideline Title:** WHO Guidelines on HIV Self-Testing and Partner Notification Services (2025 Update)
 - Publication Date: 3 March 2025
 - Target Population: All people eligible for HIV testing, key populations, adolescents
 - Scope and Objectives: Expand access to HIV testing services and improve linkage to care for undiagnosed PLHIV

5. **Guideline Title:** WHO Guidelines for HIV Prevention, Treatment and Care for Key Populations (2024 Revision)

- **Publication Date:** 10 November 2024
- **Target Population:** Key populations (sex workers, men who have sex with men, trans people, people who inject drugs, prisoners)
- **Scope and Objectives:** Address structural and access barriers to HIV care for disproportionately affected populations

1.2 **Key Recommendations**

Recommendation	GRADE Evidence Quality	Strength
All WHO HIV guideline development panels must include at least 2 voting youth representatives (10-24 years) with lived experience of HIV	Moderate	Strong
Dolutegravir-based antiretroviral therapy (ART) is recommended as first-line treatment for all PLHIV, including pregnant/breastfeeding people and children ≥ 4 weeks old	High	Strong
Routine cryptococcal antigen (CrAg) screening is required for all PLHIV with CD4 count < 100 cells/mm ³ prior to ART initiation	High	Strong
HIV self-testing is recommended as the first-choice testing option for key populations and adolescents facing barriers to facility-based testing	Moderate	Strong
Oral emtricitabine/tenofovir disoproxil fumarate (FTC/TDF) pre-exposure prophylaxis (PrEP) is recommended for all members of key populations at ongoing HIV risk	High	Strong

1.3 **Guideline Development Methodology**

- **Evidence Review Process:** Systematic search of PubMed, Embase, Cochrane Library, and WHO regional databases for studies published 2020-2026; independent appraisal of randomized controlled trials (RCTs), systematic reviews, meta-analyses, and real-world evidence by 2 separate methodological teams
- **Consensus Method:** Modified Delphi process with 3 rounds of anonymous voting, followed by in-person consensus meeting of 45 multidisciplinary panel members (clinicians, public health experts, PLHIV representatives, youth representatives, policy makers); recommendations required ≥ 80% panel agreement for adoption
- **Conflict of Interest Management:** All panel members completed WHO standardized COI disclosure forms; members with direct financial or intellectual COIs related to guideline topics were recused from voting on relevant recommendations; no industry funding was accepted for guideline development

II. **Evidence Summary After WHO Latest Guideline Release (Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline) (Evidence Cutoff Date: 29 April 2026)**

2.1 New Evidence Overview

- **Search Strategy:** Systematic search for studies published between 15 January 2025 (release date of the 2025 WHO consolidated HIV guideline) and 29 April 2026, restricted to English-language peer-reviewed publications, regulatory reports, and national public health surveillance data
- **Databases Searched:** PubMed, Embase, Cochrane Central Register of Controlled Trials, *Lancet HIV*, *NEJM Evidence*, CDC HIV Surveillance Reports, ECDC HIV Monitoring Reports
- **Inclusion Criteria:** RCTs, large-scale ($n \geq 1000$) observational studies, systematic reviews/meta-analyses focused on HIV diagnosis, treatment, prevention, or service delivery; studies with clinically relevant endpoints (mortality, virologic suppression, HIV incidence, testing uptake)
- **Exclusion Criteria:** Small single-site studies, case reports, editorials, non-peer-reviewed preprints without independent validation, non-human studies

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
INSIGHT START 2026 Extension Study	Multicenter open-label RCT (n = 16,762)	Immediate ART initiation in PLHIV with CD4 count > 500 cells/mm ³ reduced all-cause mortality by 22% compared to deferred initiation (HR 0.78, 95% CI 0.67-0.91, $p = 0.001$) over 10 years of follow-up, with no increased risk of adverse events	High
PrEPX Global Longitudinal Study	Prospective observational cohort (n = 12,473)	Long-acting injectable cabotegravir PrEP administered every 8 weeks had 94% effectiveness against HIV acquisition among trans women and men who have sex with men, compared to 86% effectiveness for daily oral PrEP ($p < 0.001$), with higher adherence rates (89% vs 72% for oral PrEP)	Moderate
WHO African Region HIV Self-Testing Implementation Trial	Cluster RCT (n = 32,109 across 6 LMICs)	Community distribution of free HIV self-test kits with digital follow-up support increased HIV testing uptake among adolescents (15-19 years) by 68% compared to standard facility-based testing, with 92% of newly diagnosed individuals linked to ART within 30 days	High
<i>Lancet HIV</i> 2026 Pediatric HIV Treatment Meta-Analysis	Systematic review and meta-analysis of 17 RCTs (n = 4,218 children < 10 years living with HIV)	Dolutegravir-based ART in children < 2 years old resulted in 31% higher virologic suppression rates at 48 weeks compared to efavirenz-based regimens (RR 1.31, 95% CI 1.19-1.44, $p < 0.001$), with no increased risk of neurodevelopmental adverse events	High

2.3 Evidence Gaps and Future Research Needs

- Limited data on long-term safety and effectiveness of long-acting ART and PrEP regimens in pregnant/breastfeeding people and children < 12 years old
- Gaps in evidence on optimal HIV care models for rural and low-resource settings with limited access to viral load and CD4 count laboratory monitoring
- Lack of standardized metrics to evaluate the impact of youth engagement in guideline development on HIV care outcomes for adolescent and youth populations
- Limited research on HIV prevention and care outcomes for disproportionately affected marginalized groups, including undocumented migrants, homeless populations, and intersex people
- Need for large-scale trials evaluating the cost-effectiveness of next-generation HIV prevention tools, including broadly neutralizing antibodies (bNAbs) and mRNA-based HIV vaccines, in LMICs

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
ART is recommended for all PLHIV, with immediate initiation strongly recommended for PLHIV with CD4 count < 350 cells/mm ³ , and conditional recommendation for immediate initiation for PLHIV with CD4 count > 350 cells/mm ³	High-quality RCT evidence demonstrates 22% reduction in all-cause mortality with immediate ART initiation regardless of baseline CD4 count, with no excess adverse events	Immediate ART initiation is strongly recommended for all PLHIV, regardless of baseline CD4 count or clinical stage (GRADE Quality: High, Strength: Strong)
Oral FTC/TDF PrEP is recommended for all people at ongoing HIV risk; long-acting injectable cabotegravir PrEP is conditionally recommended only for people who cannot adhere to daily oral PrEP	Moderate-quality large cohort evidence shows significantly higher effectiveness and adherence for long-acting injectable cabotegravir PrEP compared to oral PrEP among key populations	Long-acting injectable cabotegravir PrEP every 8 weeks is strongly recommended as a first-choice PrEP option for all people at ongoing HIV risk, in addition to oral PrEP (GRADE Quality: Moderate, Strength: Strong)
HIV self-testing is recommended as a first-choice testing option for key populations and adolescents facing barriers to facility-based testing, with conditional recommendation for community distribution of self-test kits	High-quality cluster RCT evidence demonstrates 68% higher testing uptake among adolescents, with high linkage to care, via community distribution of free self-test kits with digital follow-up	Community-wide free distribution of HIV self-test kits with digital linkage to care support is strongly recommended for all settings with low HIV testing coverage among adolescents and key populations (GRADE Quality: High, Strength: Strong)
Dolutegravir-based ART is recommended as first-line treatment for children ≥ 4 weeks old, with conditional recommendation for use in children < 2 years old due to limited safety data	High-quality meta-analysis evidence shows higher virologic suppression and no increased neurodevelopmental risk of dolutegravir-based ART in children < 2 years old	Dolutegravir-based ART is strongly recommended as first-line treatment for all children living with HIV ≥ 4 weeks old, including children < 2 years old (GRADE Quality: High, Strength: Strong)